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5pPP2. A cognitive approach for binaural models using a top-down feedback structure

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Current models to explain binaural hearing generally focus on bottom-up processes of the auditory periphery and subcortical brain functions to simulate sound localization and other binaural tasks. While these models have been very successful in explaining a number of psychoacoustic phenomena, their architecture is not suitable to simulate experiments that involve cognition. The project presented here seeks to close the gap between functional binaural models and research in applied robotics. A software architecture that was originally designed to simulate the process of music improvisation using a combination of Computational Auditory Scene Analysis, machine learning and logic-based reasoning, the Creative Artificially-Intuitive and Reasoning Agent CAIRA was extended to simulate a number of basic binaural phenomena including sound localization of multiple-sources, resolving front/back confusions through strategic head movements, and adapting inhibitory parameters to the presented signals to evoke localization dominance.

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INTRODUCTION

Current binaural models typically analyze sound using solely bottom-up processes. In the model, signals are run through a number of stages simulating the auditory periphery. These processes include (i) convolution with head-related transfer functions (HRTFs) to simulate the frequency-specific amplitude and phase distortion due to the head, pinna and torso, (ii) band-pass filtering to mimic the function of the basilar membrane, (iii) processes to simulate the non-linear behavior of hair-cells, (iv) analysis of interaural time differences (ITDs) and/or interaural level differences (ILDs) (v) and, finally, a mechanism to estimate binaural parameters, for example the location of an auditory event.

Unlike humans, these models typically do not explore their environment through active top-down mechanisms. Instead, they process any given sound using the same pattern. One example of a top-down process is the strategic use of head movements. Perrett and Noble [1] have shown that front/back errors are reduced from 30% for the no head-movement condition to approximately 1–2% for the natural condition where the human subjects are allowed to move their head in any way they want. The reduction in error was similar for the case where the subjects were instructed to perform a given head trajectory. The most important localization cues, ILDs and ITDs, are very similar for the corresponding front and back angles, e.g., angles with equal degree of lateralization, but different hemisphere (front or back). Head movements serve to help resolve these ambiguities.

Recently, a binaural mechanism was proposed that can simulate the experiments of Perret and Noble [1]. The algorithm, which will be further introduced below, can perform virtual head movement to discriminate between front and back directions [2].

The model is the third in a series to use some form of top-down feedback structure for binaural sound analysis. The other two models consist of an algorithm to remove information from concurrent sound sources to better localize a target sound source [3] and a precedence effect model that can inhibit the influence of reflections more effectively by actively adjusting specific parameters concerning the delay time and amplitude ratio between the direct sound source and a reflection [4].

The aim of this study was to investigate how the three models can be combined to design more complex multi-functional binaural models. To this purpose, the head-movement model was incorporated into these models.

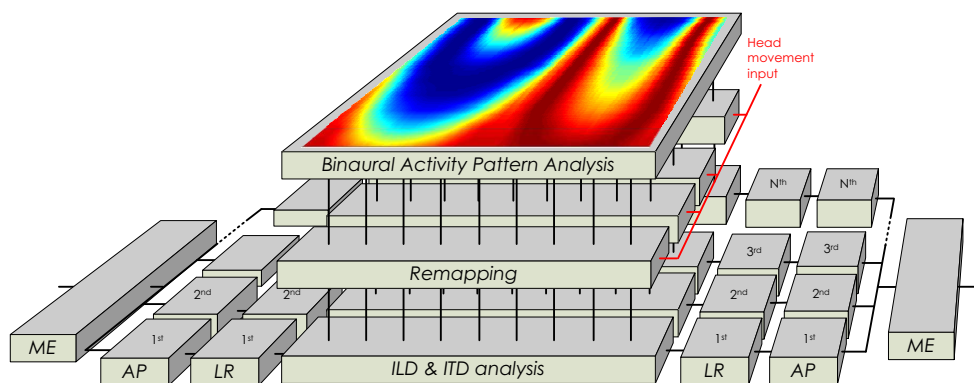


FIGURE 1: General model structure of the binaural localization model utilizing head movements.

BINAURAL MODEL WITH HEAD MOVEMENTS

General Binaural Model Structure

The head-movement model is based on an interaural cross-correlation algorithm. Stages to simulate the auditory periphery are implemented as shown in Fig. 1. This model is similar to the one proposed by Blauert and Cobben [5] and Braasch [3]. The transformations from the sound sources to the eardrums due to the influence of the outer ear and, occasionally, room reflections are taken into account by filtering the sounds with HRTFs from a specific direction. Afterwards, the outputs for all sound sources are added together for the left and right channels. Basilar-membrane and hair-cell behavior are simulated with a gammatone filter bank with 36 bands at a sampling frequency of 48 kHz, as described by Patterson and colleagues [6], and with a simple half-wave rectification. Only the frequency bands 1 to 16 are analyzed to take into account that the human auditory system cannot resolve the temporal fine structure at high frequencies, as well as the fact that the time differences in the fine structure of the lower frequencies are dominant – if they are available [7]. After the half-wave rectification, the interaural cross correlation (ICC):

$$\Psi_{y_l,r}(t, \tau) = \int_{t'=t}^{t+\Delta t} y_l(t' - \tau/2) \cdot y_r(t' + \tau/2) dt', \quad (1)$$

is performed with the internal delay τ , the start time t , the integration time Δt , and the left and right sound pressure signals $y_l(t)$ and $y_r(t)$. The interaural cross correlation is calculated every 5 ms within each frequency band.

Head-movement algorithm

For the proposed model, it is important to remap the axis of the cross correlation functions from interaural time differences to azimuth positions. Otherwise, the peaks of the cross-correlation functions will not necessarily line up at one lag for a single sound source because the ITDs of the HRTFs are frequency dependent. To calculate the ITDs of the HRTFs throughout the horizontal plane, the HRTF catalog, measured in a resolution of 15° in the horizontal plane, is interpolated to 1° resolution using the spherical spline method. After filtering the HRTFs with the gammatone filter bank, the ITDs for each frequency band and angle are estimated using the interaural cross-correlation algorithm. This frequency-dependent relationship between ITDs and azimuthal angles is used to remap the output of the cross-correlation stage (ICC curves) from a basis of ITDs to a basis of azimuth angles in every frequency band.

In the decision device, the average of the remapped interaural cross correlation functions over the frequency bands 1–16 is calculated and divided by the number of frequency bands. The model estimates the sound sources at the positions of the local peaks of the averaged interaural cross correlation function.

Now, in a hypothetical experiment, we can observe the following: If the sound source is in the front, the auditory event shifts right within the head-related coordinate system if the head moves left. If the sound source is in the rear, the auditory event shifts left if the head moves into the same direction. We will use this observation to eliminate the secondary peak that does not belong to the actual sound-source position. This will be done by counter-compensating the head-movement by rotating the remapping function in the opposite direction of the head movement – see [2] for further details.

A few notes need to be made concerning the temporal processing of the sound stimuli. Each test stimulus was a broadband noise burst, at constant location in an anechoic room. The stimulus was segmented in time using a running hanning window. For each time segment, HRTFs

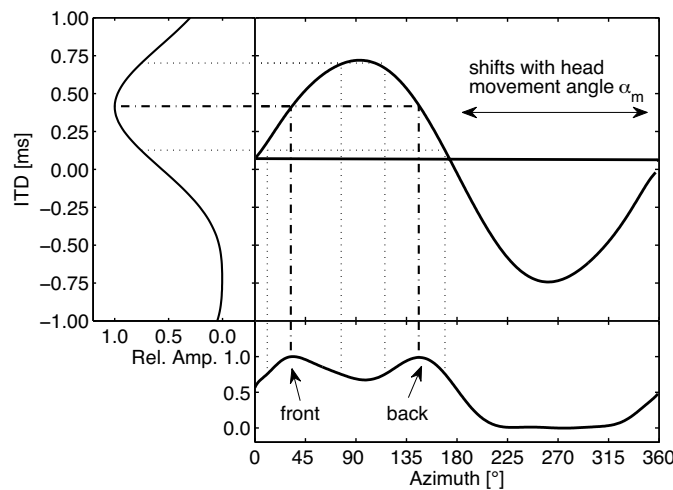


FIGURE 2: Remapping of the cross-correlation function from ITD to azimuth angle, shown for the frequency band 8 centered at 434 Hz. The signal was presented at 30° azimuth, 0° elevation.

with the actual updated head-related azimuth angle are convolved with the signal. Head related azimuth angles are computed from room-related azimuth angles and known head positions. It is also important to note that the head positions were predetermined as trajectory and known to the model.

Figure 2 shows an example of a sound source in the horizontal plane with an azimuth of 30° for the eighth frequency band. The top-left graph shows the original interaural cross correlation curve obtained using Eq. 1 as a function of ITD. The graph is rotated by 90° with the interaural cross correlation on the x-axis and ITD on the y-axis to demonstrate the remapping procedure. The curve has only one peak at an ITD of 0.45 ms. The top-right graph depicts the relationship between ITD and azimuth for this frequency band. As mentioned previously, the data was obtained by analyzing HRTFs from a human subject. Now, we use this curve to project every data point of the ICC vs. ITD function to an ICC vs. azimuth function as shown for a few data points using the straight dotted and dashed-dotted lines. The bottom panel shows the remapped interaural cross correlation function, which now contains two peaks – one for the frontal hemisphere and one for the rear hemisphere. The two peaks fall together with the points where the cone-of-confusion hyperbolas intersect the horizontal plane for the ITD value of the maximum peak that is shown in the top-left panel.

Results

Figure 3 shows the averaged interaural cross correlation curve for a bandpass-filtered white noise signal with a duration of 70 ms. The curve is shown for every 5-ms time step. It is averaged over time from the beginning of the stimulus to the time as indicated by the time axis. The signal is positioned at 30° azimuth in the room-related coordinate system. At the beginning of the stimulus presentation, the head is oriented toward the front ($\alpha_h=0$) and then rotates with a constant angular velocity to the left until it reaches an angle of 90° when the stimulus is turned off. In the beginning of the simulation both peaks for the front and back location have equal heights, but over time the peak for the rear location smears out and diminishes further. The solid black line shows the position of the true location in the front, which remains stable over time, the dashed black curve shows the maximum peak position for the confused rear position.

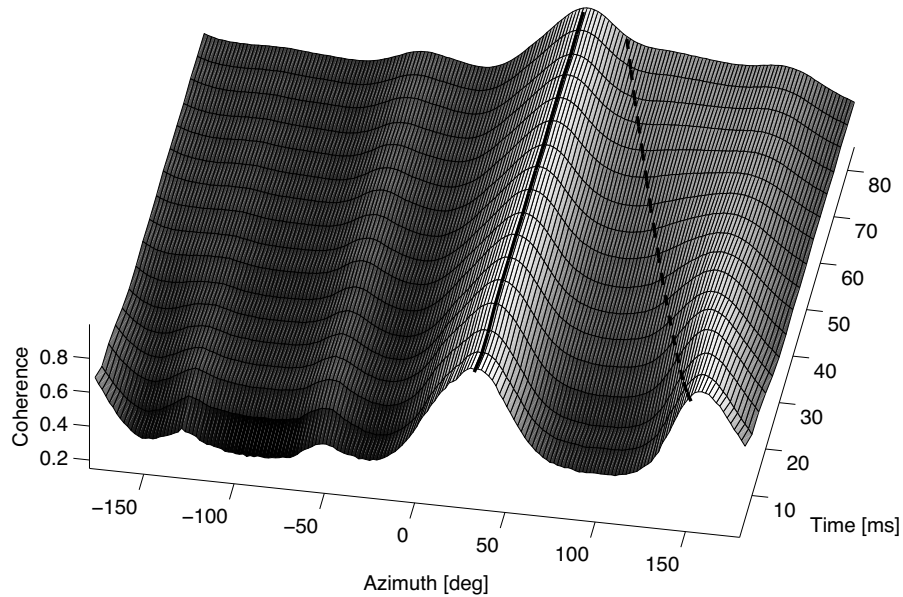


FIGURE 3: Interaural cross-correlation pattern for a sound source at 30° that is presented during a head rotation from $\alpha_m=0^\circ$ to 45° . The averaged interaural cross correlation functions are shown for every time step of 5 ms. The solid black line shows the azimuth of the sound source $\alpha_r=30^\circ$. The dashed black curve depicts the momentary location of the secondary peak that dissolves over time because of the head movement.

Integration of the Interaural Cross-Correlation Difference Model

The Interaural Cross-Correlation Difference (ICCD) model was developed to simulate the human ability to correctly localize an ongoing broadband target sound in the presence of a broadband masker [3, 8]. Since the masker and target are assumed to be uncorrelated signals, the cross terms are zero. Therefore the model takes the interaural cross correlation function of the total signal, $\psi_Y(f, \tau)$ to be the sum of the interaural cross correlation function of the masker/distracter, $\psi_D(f, \tau)$, and the interaural cross correlation function of the target $\psi_T(f, \tau)$:

$$\psi_Y(f, \tau) = \psi_T(f, \tau) + \psi_D(f, \tau) \quad (2)$$

For the classic test paradigm, where the masker precedes but fully overlaps the target in time, the model substitutes the interaural cross correlation of the masker in the total signal, $\psi_D(f, \tau)$, with the interaural cross correlation of the preceding masker part, $\psi_{D'}(f, \tau)$. Using this approach the interaural cross correlation of the target can be estimated:

$$\psi_T(f, \tau) = \psi_Y(f, \tau) - \psi_D(f, \tau) \quad (3)$$

$$\approx \psi_Y(f, \tau) - \psi_{D'}(f, \tau). \quad (4)$$

While in the original paper [3], the interaural cross correlation is calculated over the whole target and preceding masker durations, the model had to be extended for this study. Now the interaural cross correlation is analyzed in steps of 5 ms. After every step the interaural cross correlation is remapped for every frequency band. Head movements are included and compensated for as described in the section *Head-movement algorithm*.

The signals had a duration of 140 ms. During the first 70 ms, the masker was presented in isolation in the second 70 ms in presence of the target. The head of the model was turned virtually from -32° to 32° during the stimulus presentation (preceding masker from -32° to 0° , target+masker from 0° to 32°).

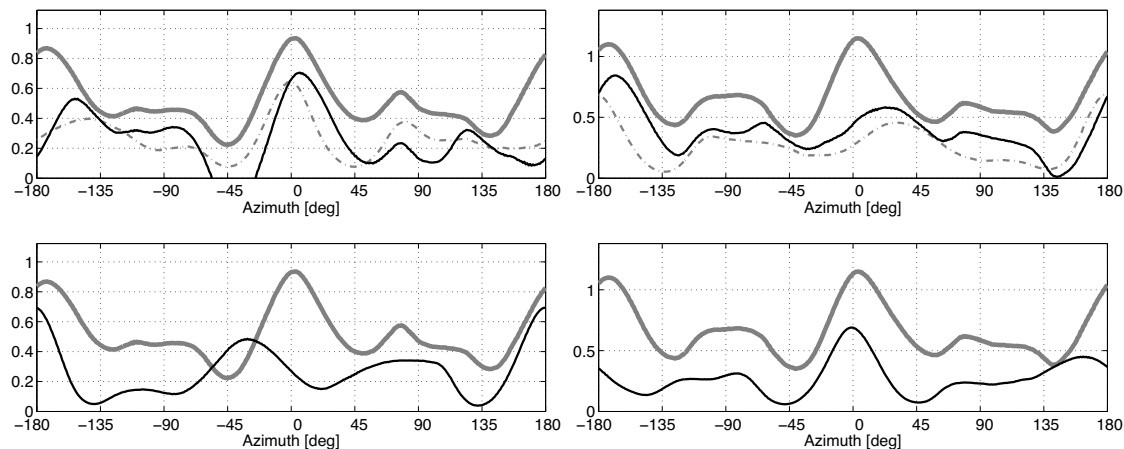


FIGURE 4: Localization results for the combined ICCD and head-movement model utilizing head movements. Left: target 0° azimuth, 180° masker; right: target 180° azimuth, 0° masker.

Figure 4 shows the model results. The top-left graph shows the condition for which the target sound is located at 0° and the masker is positioned at 180° . The solid grey curve shows the interaural cross correlation function of the masker/target pair. Even though the model moves its head, both peaks at 0° and 180° maintain equal height. However, for the preceding masker part, the model correctly eliminates the secondary peak at 0° (bottom left graph, solid black curve). Now if we subtract this curve from the total signal (solid grey curves in the top and bottom left graphs), the model can correctly predict the front/back direction of the masker (top-left graph, solid black curve). Only the peak at the correct position (0°) prevails, while the secondary peak in the rear is washed out. The interaural cross correlation for the isolated target signal is shown by the grey dash-dotted line for comparison purposes. The top-right graph of the same figure shows the condition in which the masker and target positions are swapped. The cross-correlation function for the total signal (grey curves) is very similar, but the peak of the preceding masker is now located at 0° . Subtracting it from the total cross-correlation function correctly identifies the target location at 180° .

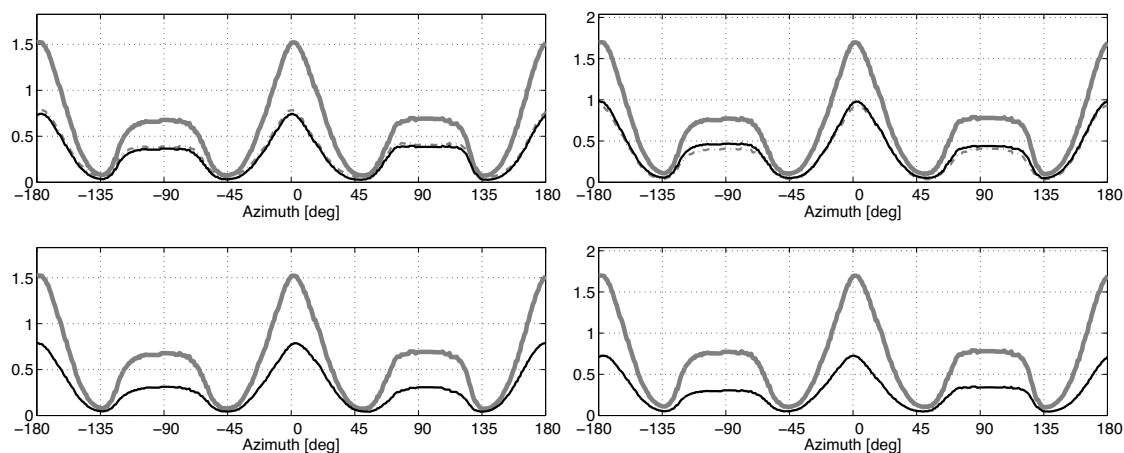


FIGURE 5: Same as Fig. 4 but without head movements.

The two graphs in Fig. 4 show the same condition but without head-movements. Note that the correlation peaks of the total signal (solid grey curve) are higher for both no head-movement cases than for the two head-movement cases, because they have not been partly washed out through the head movements. Since the peaks for the front/back direction of the preceding masker are equally high (bottom graphs, solid black curves), the model can no longer predict the correct target directions (solid black curves, top graphs). That is, while the model can estimate the correct left/right positions, it cannot determine the front/back direction correctly as both interaural cross correlation peaks have equal heights without the use of head movements.

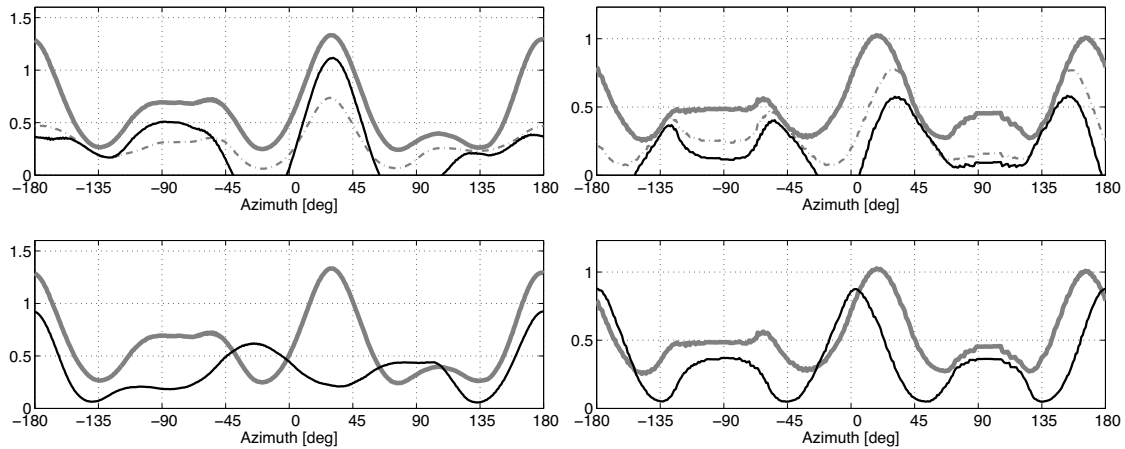


FIGURE 6: Localization results for the combined ICCD and head movement model with head movements. Left: target 30° azimuth, 180° masker; right: the same condition but without head movements

In the third condition, the target is located outside the median plane (30°) with the masker location at 180° (Figure 6). Here it is noteworthy that the model can resolve the front/back directions for masker and target (top-left graph, solid grey curve). The ICCD algorithm also extracts the correct target peak at 30°. However, without the ICCD algorithm, the model cannot determine which of the signals is masker and which one is target. In the no head-movement case, the cross-correlation functions of masker and target combine to a single peak for the front and back directions (top-right graph, solid grey curve).

Integration of the Precedence Effect Model

Our second model extension focuses on the precedence effect in connection with head movements. Very little is known about how the precedence effect prevails if a listener turns the head, changing the parameters between the direct and the reflected sounds. In this study, we examine whether a precedence model can handle this situation when a head-movement compensation algorithm is used as described in Section *Head-movement algorithm*.

A recently introduced binaural model [4] is used to simulate the precedence effect. The model estimates the optimal parameters to eliminate a single reflection, the lag, using an auto-correlation algorithm by estimating the delay and amplitude of the reflection:

$$s_t(t) = s_d(t) + s_r(t) = s_d(t) + r \cdot s_d(t - T). \quad (5)$$

with the total signal $s_t(t)$, the direct signal $s_d(t)$ (the lead), the delay time T , the Lead/Lag Amplitude Ratio (LLAR) r . After the parameter estimation, an inverse filter is used to eliminate the lag signal, $s_r(t)$:

$$h_d = \sum_{n=0}^N (-r)^n \delta(T \cdot n). \quad (6)$$

This procedure is done separately for the left and right signals. The signals are processed for individual frequency bands as described in [4]. Then the signals are analyzed using the binaural model with head-movement compensation that was introduced in Section *Binaural model with head movements*.

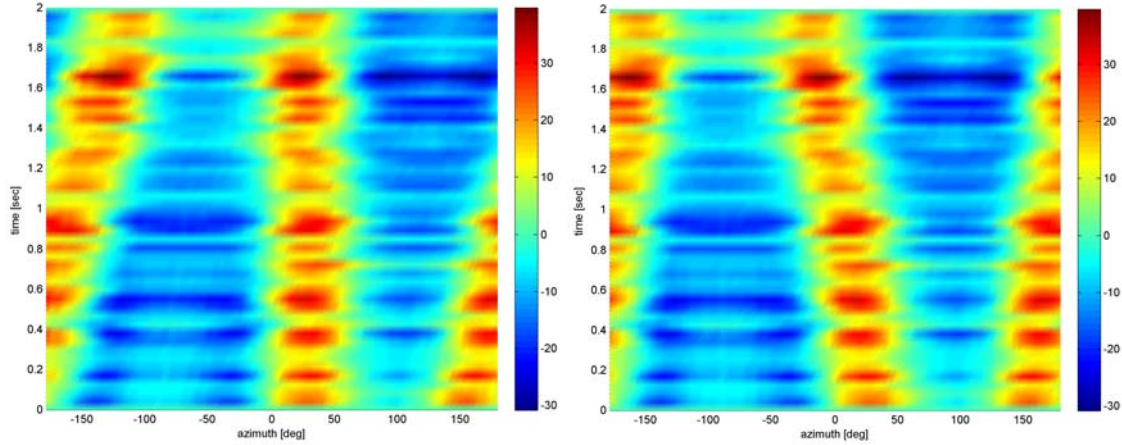


FIGURE 7: Results for a precedence effect model in combining with the head-movement compensation model. Left graph: with head-movement compensation, right graph: without head-movement compensation.

Figure 7 shows the results of the model analysis. The left graph shows the model performance without head-movement compensation, the right graph with head-movement compensation. Now the model stably maintains the peak at the correct position at 30° azimuth in the room-related coordinate system, while the secondary peak keeps moving and will wash out in a long term average.

CONCLUSIONS

It has been shown that a model capable of simulating head movements can be integrated into two extended binaural models to simulate listening task. Using this model, front/back confusion can be resolved, even if the target sound is presented with a masker or in presence of a reflection.

ACKNOWLEDGMENT

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